THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SOCIAL SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 2: **Understanding the Weather** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A station has mean daily maximum 34 degrees C and mean minimum 20 degrees C. Its mean daily range is 1

(a) 14 degrees C **(b)** 7 degrees C **(c)** 27 degrees C **(d)** 54 degrees C

2. Which instrument combination is needed to predict possible fog? 1

(a) anemometer and ruler **(b)** thermometer and hygrometer

(c) barometer and ruler **(d)** rain gauge and wind vane

3. If the daily range at a place is 2 degrees C, the place is likely 1

(a) cloudy, humid or coastal **(b)** desert

(c) polar **(d)** high mountain

4. A coastal town and an inland town both have a mean of 28 degrees C. The coastal town likely has 1

(a) a larger range **(b)** a smaller range

(c) the same range always **(d)** no range

5. Why might a cloudy night be warmer than a clear night? 1

(a) clouds trap outgoing heat **(b)** clouds remove heat

(c) clouds reflect cold **(d)** clouds create wind

6. A drop in temperature of about 6.5 degrees C per 1,000 m height means a hill station at 2,000 m is cooler than the plain by about 1

(a) 13 degrees C **(b)** 2 degrees C **(c)** 26 degrees C **(d)** 6.5 degrees C

7. A barometer reading falls from 1010 to 995 millibars in a day. This most likely means 1

(a) fair weather **(b)** no change

(c) a storm is approaching **(d)** a flood

8. If the pressure over land is lower than over the sea, the wind will blow 1

(a) upwards only **(b)** downwards only **(c)** from sea to land **(d)** from land to sea

9. **Assertion (A):** Wind blows from high-pressure areas to low-pressure areas. 1

Reason (R): Air tends to move to balance a difference in pressure.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** Mean daily temperature is calculated as (maximum + minimum) / 2. 1
Reason (R): Rainfall is measured with a rain gauge.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are daily weather records useful over years? 2

12. Why is Earth heated unevenly? 2

13. Why is forecasting more difficult in the tropics? 2

14. Why do isobars close together mean strong wind? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A town recorded max/min as 36/24, 38/26 and 34/22 on three days. Find mean temperature and range of each day. 3

16. Interpret a weather map with closely spaced isobars. 3

17. Evaluate how cyclone warnings help coastal communities. 3

OR

Analyse why the humidity and temperature together affect comfort. 3

18. Assess the role of satellites in modern forecasting. 3

OR

Examine why daily forecasts differ in accuracy from seasonal forecasts. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A school weather club in Jaipur records the temperature each day. On one day the maximum temperature was 38 degrees C and the minimum was 24 degrees C. The students also fix a rain gauge in the playground, which collected 16 mm of rain after a storm. 5
(i) Calculate the mean daily temperature.
(ii) What does the rain gauge measure?
(iii) Name the instrument used to measure temperature.
(iv) Why do weather stations keep records of the maximum and minimum temperature?

20. Analyse how pressure, wind and humidity interact to produce rainfall. 5

OR

Explain the importance of the India Meteorological Department. 5

END OF QUESTION PAPER Total: 40 Marks